Principles of Organic Chemistry

1 2 3 4 bonds

HONC

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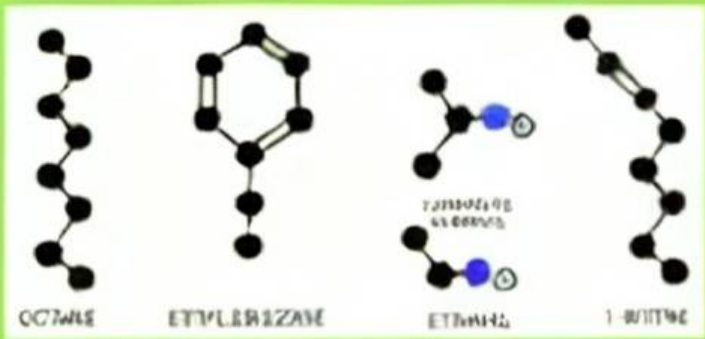
What is organic chemistry?

- study of carbon, the compounds it makes, and the reactions it undergoes
- over 16 million carbon-containing compounds are known
- because the C-C single bond (348 kJ mol^{-1}) and the C-H bond (412 kJ mol^{-1}) are strong, carbon compounds are stable
- carbon chains, rings and rings



2.1
Classes of Hydrocarbons

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Hydrocarbons

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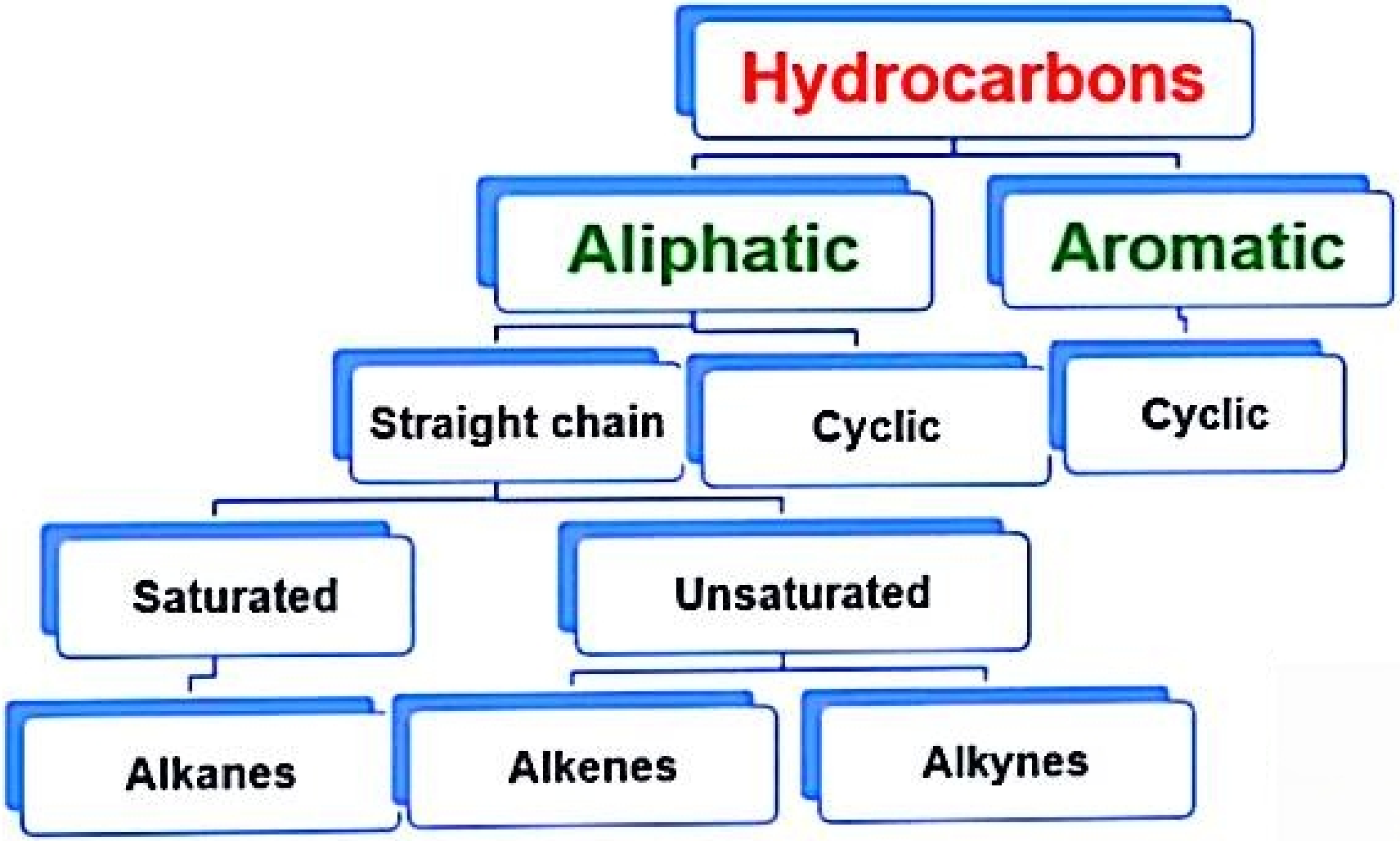

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General Molecular Formula of Hydrocarbons

1) **Alkanes:** C_nH_{2n+1} (saturated)

   i) **Cycloalkanes:** C_nH_{2n} (containing a single ring)

   ii) Alkanes and cycloalkanes are so similar that many of their properties can be considered side by side.

2) **Alkenes:** C_nH_{2n} (containing one double bond)

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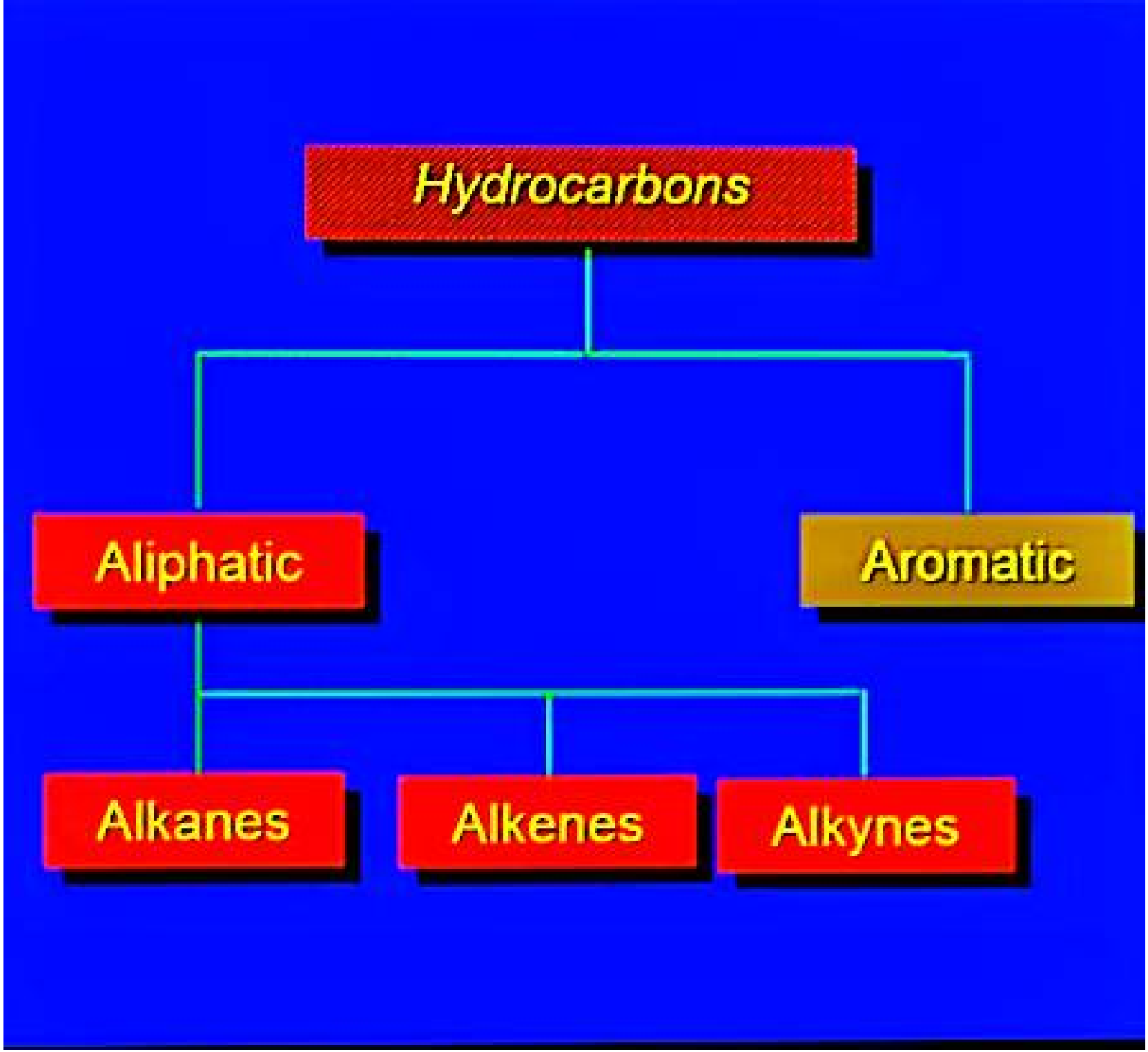

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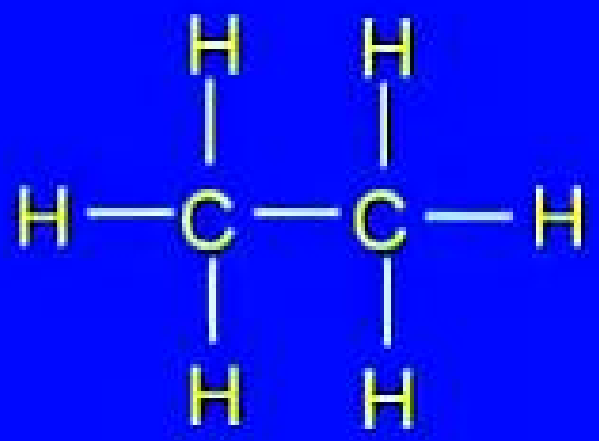
Hydrocarbons

Aliphatic

Alkanes

Alkanes are hydrocarbons in which all of the bonds are *single* bonds.



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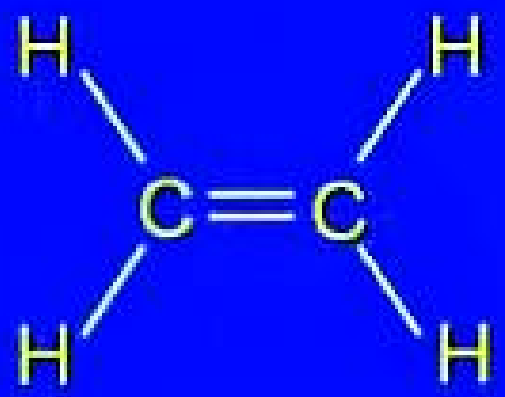
Hydrocarbons

Aliphatic

Alkenes

Alkenes are hydrocarbons that contain a carbon-carbon *double* bond.



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Hydrocarbons

Aliphatic

Alkynes are hydrocarbons that contain a carbon-carbon *triple* bond.

Alkynes

$HC \equiv CH$

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Hydrocarbons

The most common aromatic hydrocarbons are those that contain a benzene ring.

Aromatic

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